



How Stretchable Electronics Is Getting Created

Understanding of nano-scale phenomena, functional materials and soft devices has progressed to a point where substantial advances in stretchable electronics applications are becoming realistic



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Electronic devices in their various designs and capacities have already brought a revolution by influencing each and every aspect of our life. There is day-to-day dependence on technology, and the field of electronics is leading the way towards more advanced appliances.

Along with the development of electronics with various characteristics for different applications, stretchable electronics is emerging as a promising new technology for the next generation. Stretchable electronics is a technology for building electronic devices by depositing stretchable electronic circuits onto stretchable

substrates, or embedding them completely in a stretchable material such as silicone or polyurethane.

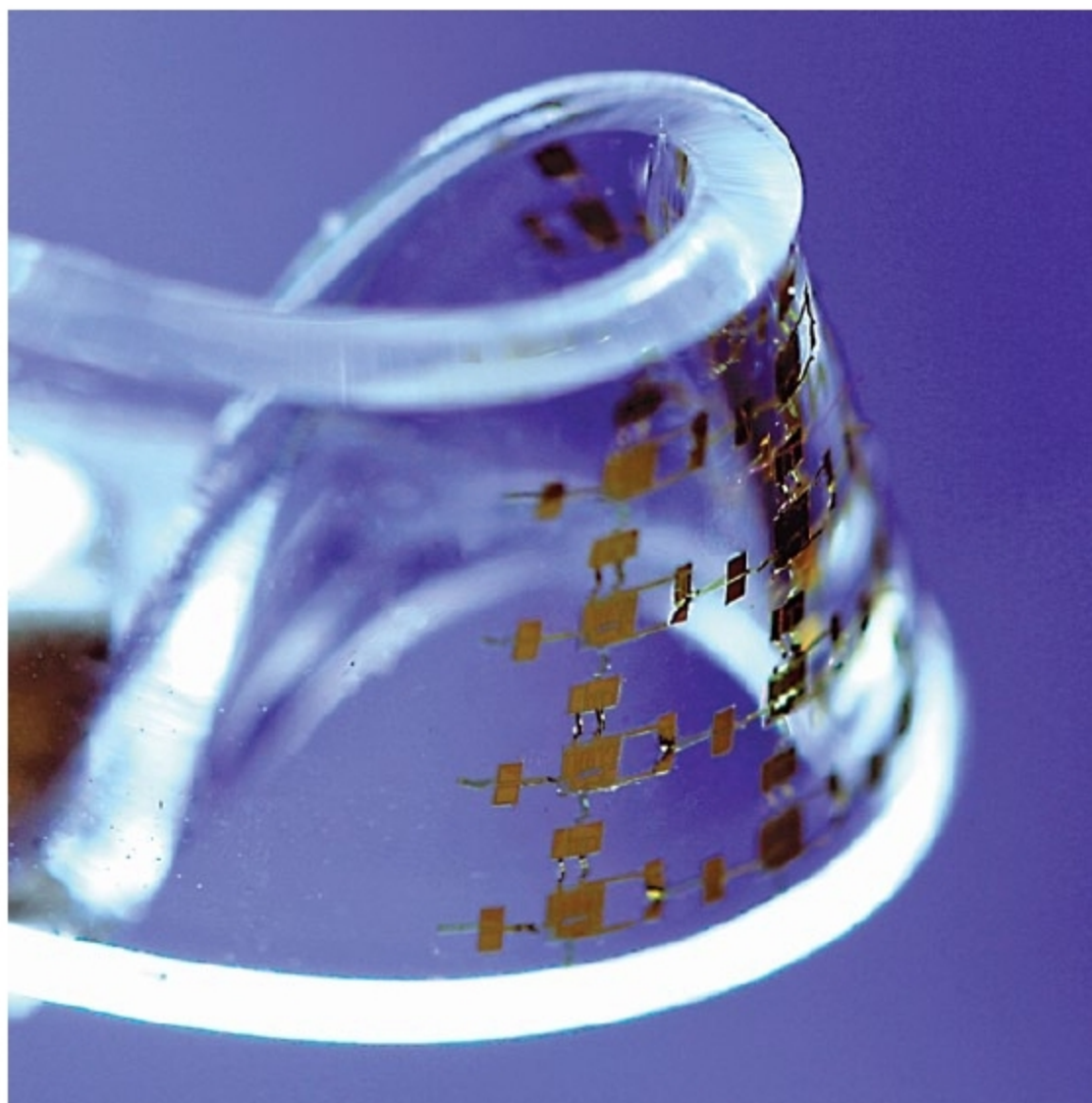
Stretchable electronics, sometimes called elastronics, is a new and emerging class of electronics that is expected to enable a range of new applications, including cyber skin for robotic devices, imparting a network of sensors on a fully conformable, stretchable cyber skin; in-vivo implantable sponge-like electronics; and flesh-like devices with embedded electronic nervous systems.

Use of wearable electronics for personal consumption and long-term remote health monitoring is growing, and the key challenge is that the system must accommodate all types of bending, curved, soft and elastic surfaces of biological objects.

Flexible and stretchable electronics is achieving a great variety of applications, because the components can be compressed, twisted and made to conform to complex non-planar surfaces. Currently, wearable electronics applications have positive effects on various aspects of our daily life, leading to economic growth, and rapid development of stretchable electronic devices and related manufacturing technologies.

Flexible, soft and stretchable forms of electronic devices enable next-generation wearable electronic applications, opening up various avenues for healthcare, energy and military purposes. Stretchable electronics is emerging in different fields such as energy conversion and storage (including solar cells), supercapacitors, lithium-ion batteries, zinc-based batteries and nanogenerators.

Further, adding transparent properties to stretchable electronic devices will make it possible to obtain more applications, such as stretchable transparent electrodes (STEs) and transparent and stretchable electronic devices (TSEDs), where high levels of both optical transparency and stretchability are



Stretchable electronics (Credit: www.azom.com)